

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

AMTI – NMTC – **25th October, 2025** – JUNIOR FINAL

Timings: 10am to 1pm

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Instructions:

1. There are 8 questions, answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your **Rough Work** should be in the answer sheet itself.

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1. $m > 0, n > 0$ are integers. Find all pairs (m, n) such that $m^2 + 3n$ and $n^2 + 3m$ are both perfect squares simultaneously.
 2. PQR is a right triangle with $\angle QPR = 90^\circ$. I is the incenter of the ΔPQR . The incircle touches side PQ at M and side PR at N. A line is drawn through I to cut PQ at A and PR at B. Prove that $PA \cdot PB \geq 4r^2$, where r is the inradius of the ΔPQR .
 3. Solve for x, y, z from the following simultaneous equations:
$$\begin{aligned}x^2 + xy + xz - x &= 7; \\y^2 + yz + yx - y &= -14; \\z^2 + zx + zy - z &= 49.\end{aligned}$$
 4. Twenty-eight integers are chosen from the interval $[104, 208]$. Prove that there exist two of them with a common prime divisor.
 5. In the parallelogram ABCD, the diagonal AC is perpendicular to the side AD. AH is drawn perpendicular to CD. The tangent at D to the circumcircle of ΔADB meets CA produced at P. Prove that: $\angle PBA = \angle DBH$.
 6. There are 6 friends, Manish, Cathy, Sundar, Elisha, Lakshaman and Divya. Manish told that the sum of our ages put together is five times my age. He again told that when Sundar will be three times my present age, the sum of my age and Divya's age will be equal to the sum of the present ages of the five of us; Elisha's age will be three times her present age and Lakshaman's age will be twice Sundar's present age plus one year. They all started working to find their present ages and Cathy asked Manish that could he tell something more. Then Manish said that his age is an odd number. How old are Manish and Sundar?
 7. Solve in positive integers: $\frac{x^2 - y}{34x - y^2} = \frac{y}{x}$, provided $34x - y^2 \neq 0$.
 8. Let ABC be an acute-angled triangle. Let points O, H be its circumcenter and orthocenter respectively. The altitude from A, the perpendicular bisector of side AC and side BC are concurrent. Then find CH:BO.

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